



# CS 498: Paged Attention

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Materials with thanks to Minjia Zhang, Arvind Krishnamurthy,  
and many colleagues

# Recall: Continuous batching and Chunked-Prefill



**Continuous  
Batching**  
(Orca)

| $T_1$ | $T_2$ | $T_3$ | $T_4$ | $T_5$ | $T_6$ | $T_7$ | $T_8$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| $S_1$ | $S_1$ | $S_1$ | $S_1$ | $S_1$ | END   | $S_6$ | $S_6$ |
| $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | END   |
| $S_3$ | $S_3$ | $S_3$ | $S_3$ | END   | $S_5$ | $S_5$ | $S_5$ |
| $S_4$ | $S_4$ | $S_4$ | $S_4$ | $S_4$ | $S_4$ | END   | $S_7$ |

# Recall: Continuous batching and Chunked-Prefill

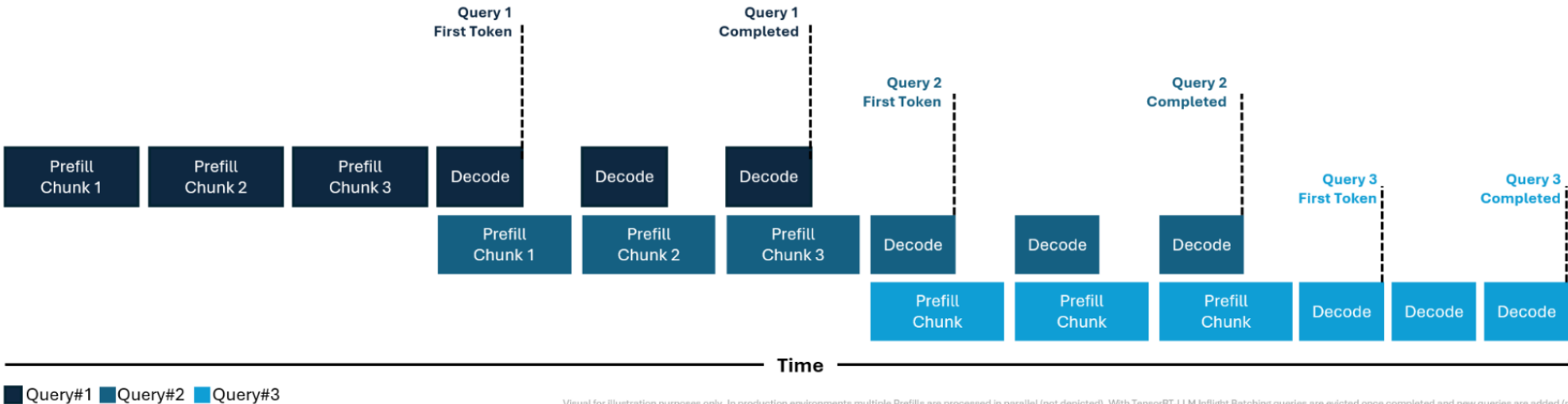


Continuous  
Batching  
(Orca)

| $T_1$ | $T_2$ | $T_3$ | $T_4$ | $T_5$ | $T_6$ | $T_7$ | $T_8$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| $S_1$ | $S_1$ | $S_1$ | $S_1$ | $S_1$ | END   | $S_6$ | $S_6$ |
| $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | $S_2$ | END   |
| $S_3$ | $S_3$ | $S_3$ | $S_3$ | END   | $S_5$ | $S_5$ | $S_5$ |
| $S_4$ | $S_4$ | $S_4$ | $S_4$ | $S_4$ | $S_4$ | END   | $S_7$ |

Chunked-  
Prefill

## W/ TensorRT-LLM Chunked Prefill



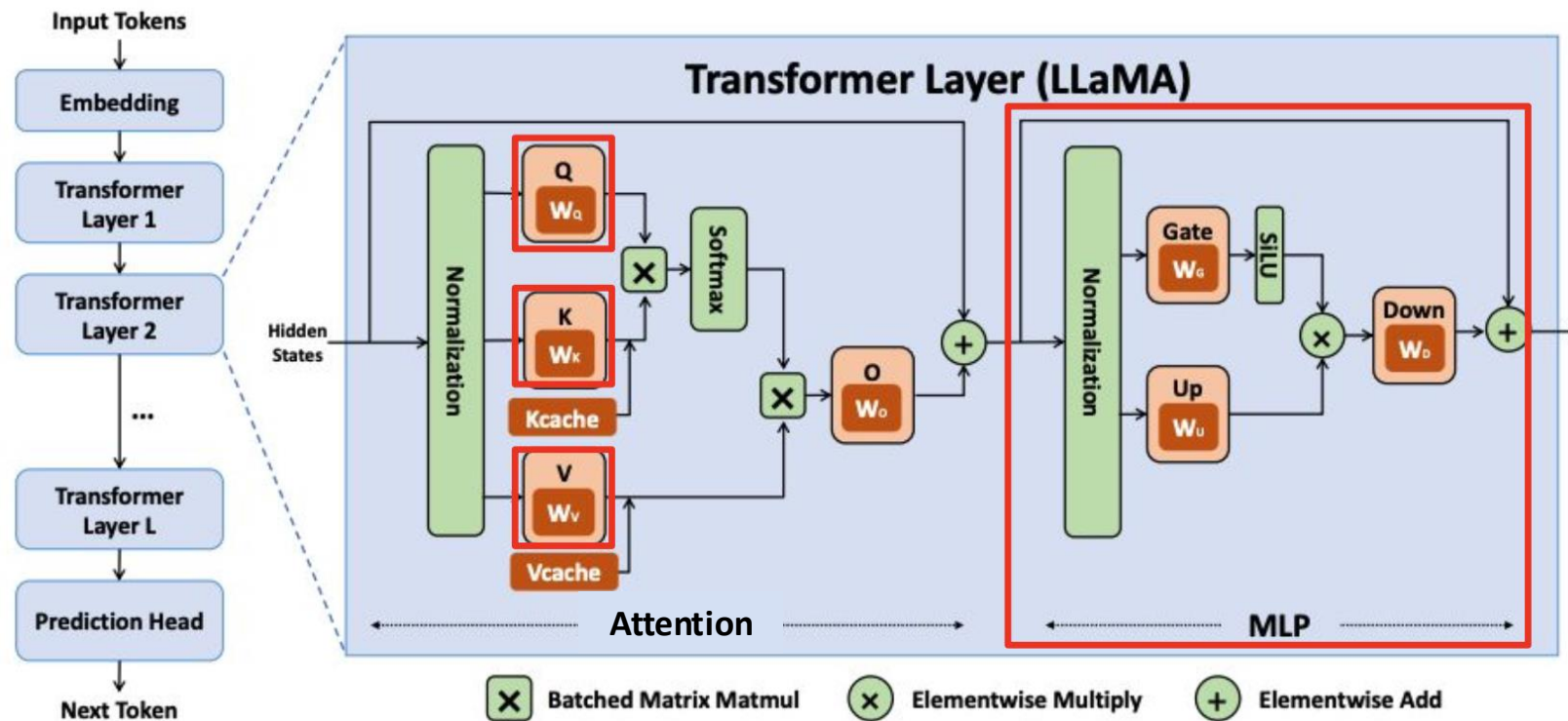
Visual for illustration purposes only. In production environments multiple Prefills are processed in parallel (not depicted). With TensorRT-LLM Inflight Batching queries are evicted once completed and new queries are added (not d

## LLM Inference with Large Batch Sizes

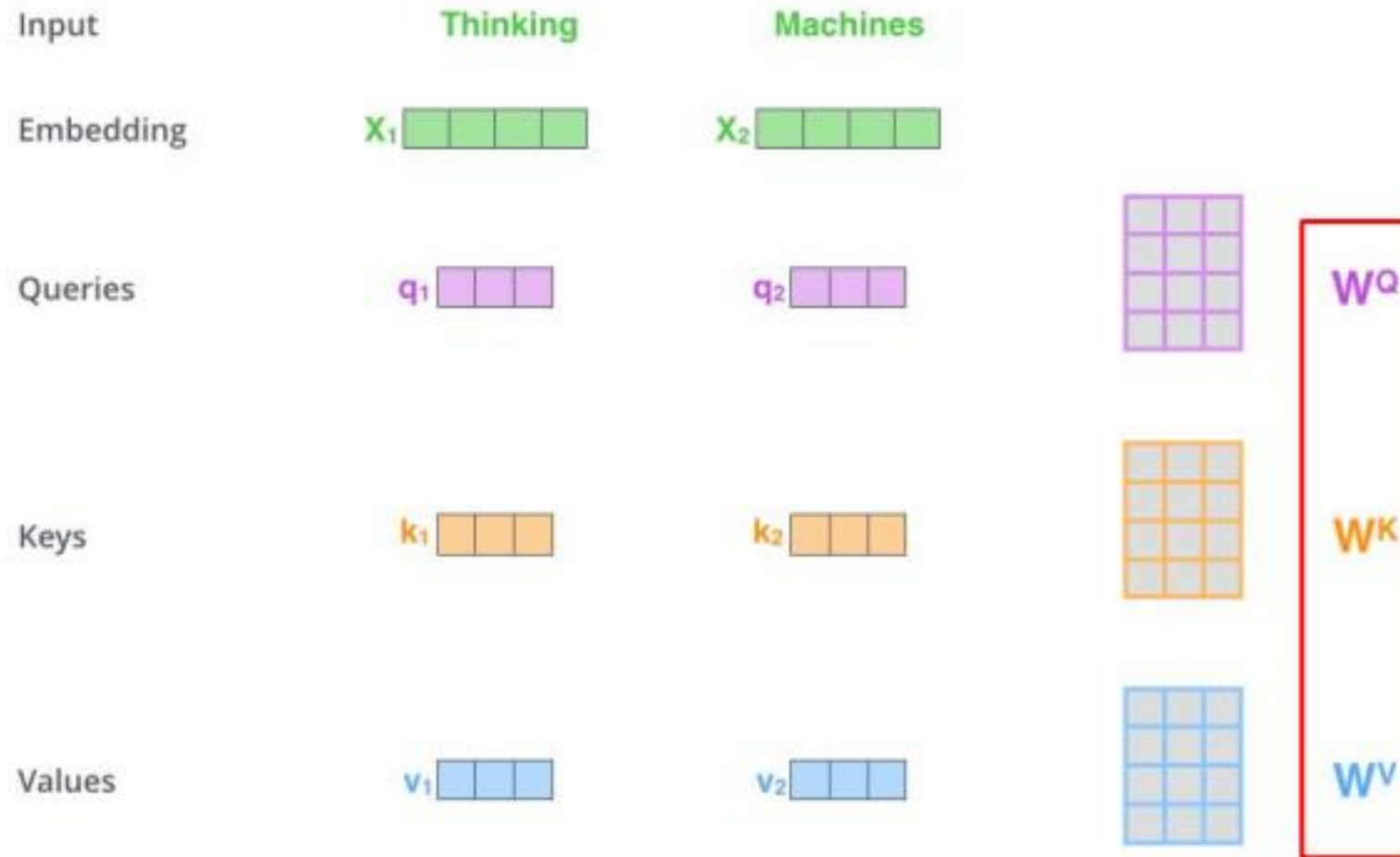
- Paged Attention

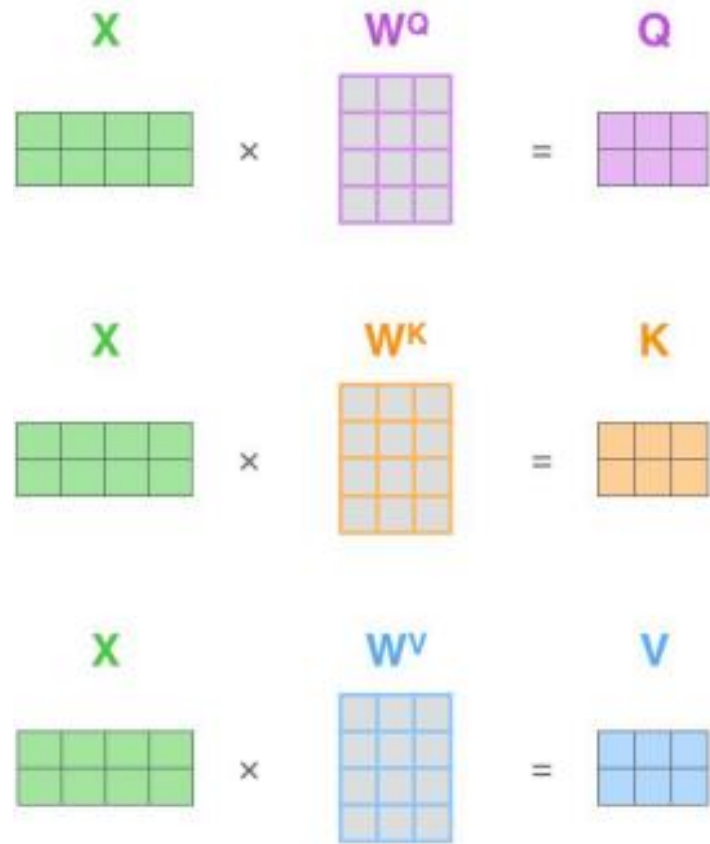
Objective: Learn why KV cache is important for LLM inference and how to manage KV cache efficiently

# Self-Attention in Transformers

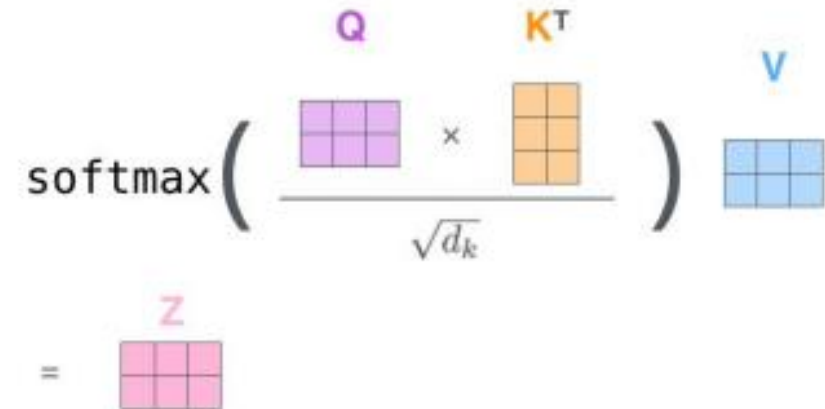


# Self-Attention





$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}$$

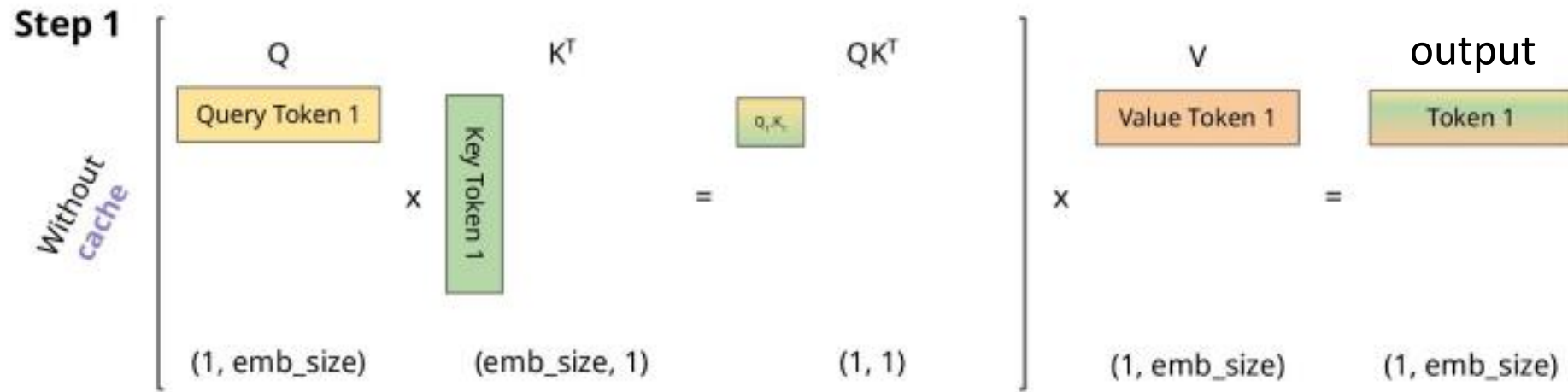


The diagram illustrates the calculation of the attention matrix Z:

- The Query matrix Q (purple 2x3) is multiplied by the transpose of the Key matrix  $K^T$  (orange 3x2).
- The result is divided by  $\sqrt{d_k}$ .
- The softmax function is applied to the result.
- The final result is multiplied by the Value matrix V (blue 2x3) to produce the attention matrix Z (pink 2x3).

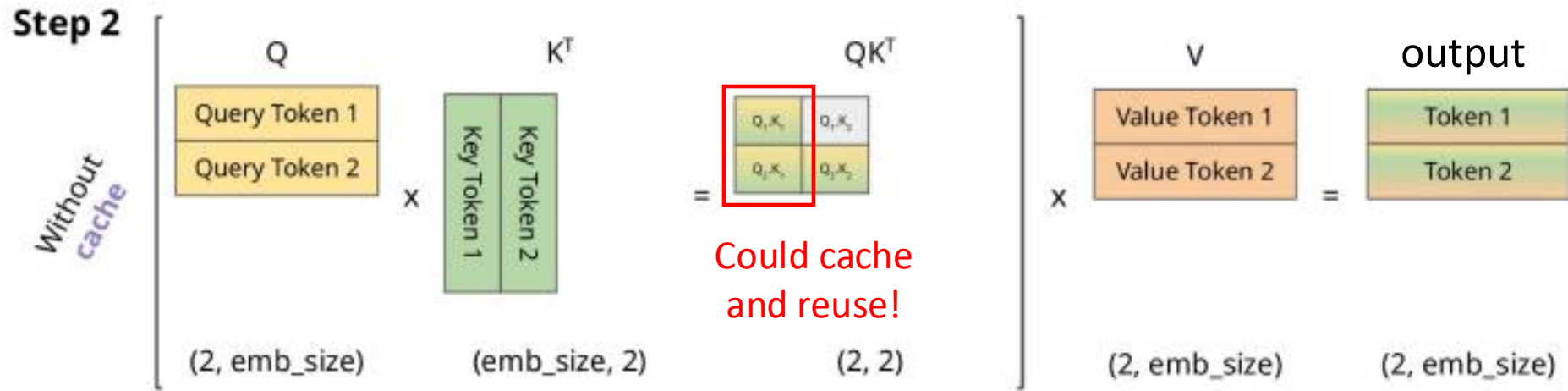


# Self-Attention with KV Cache



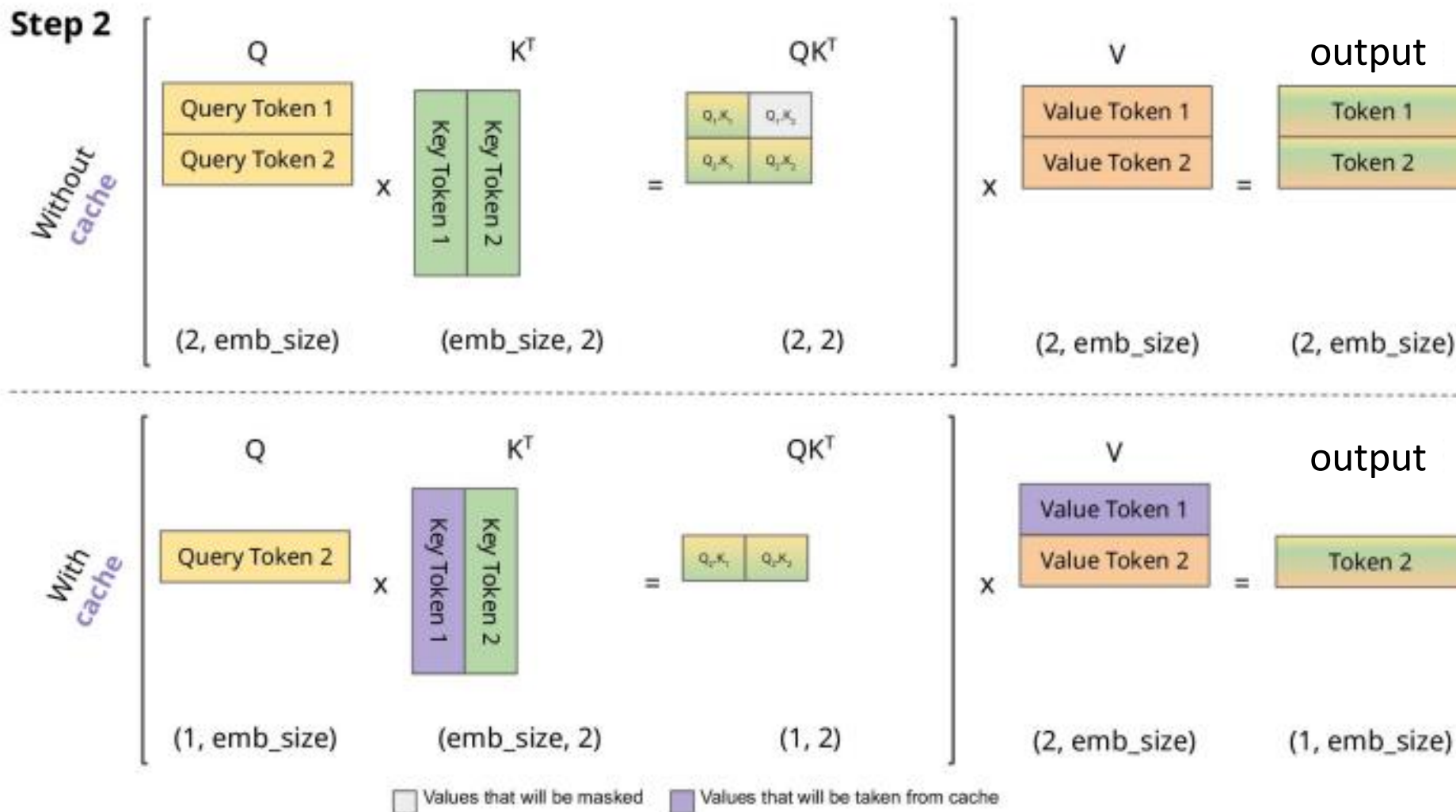


# Self-Attention with KV Cache

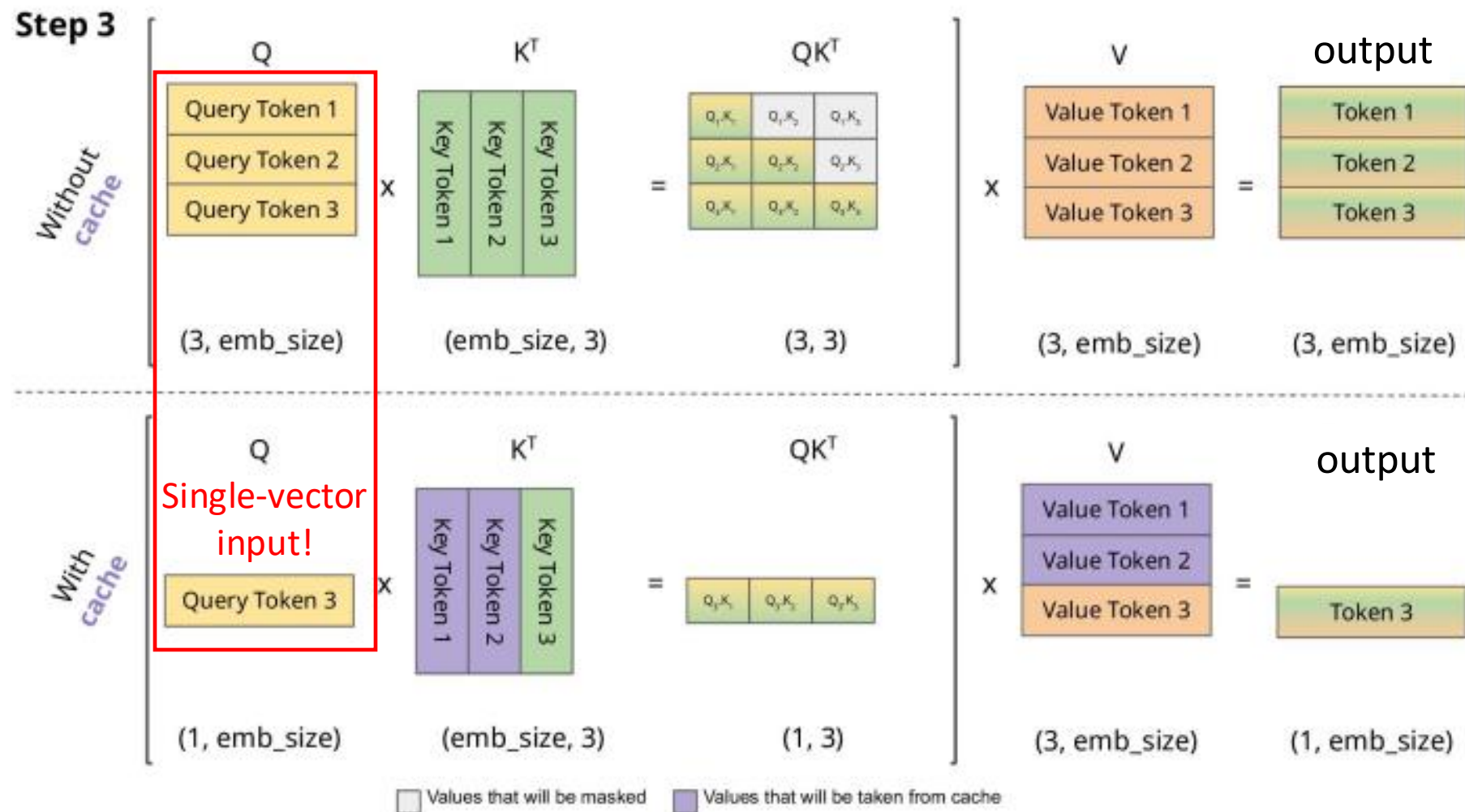


□ Values that will be masked    □ Values that will be taken from cache

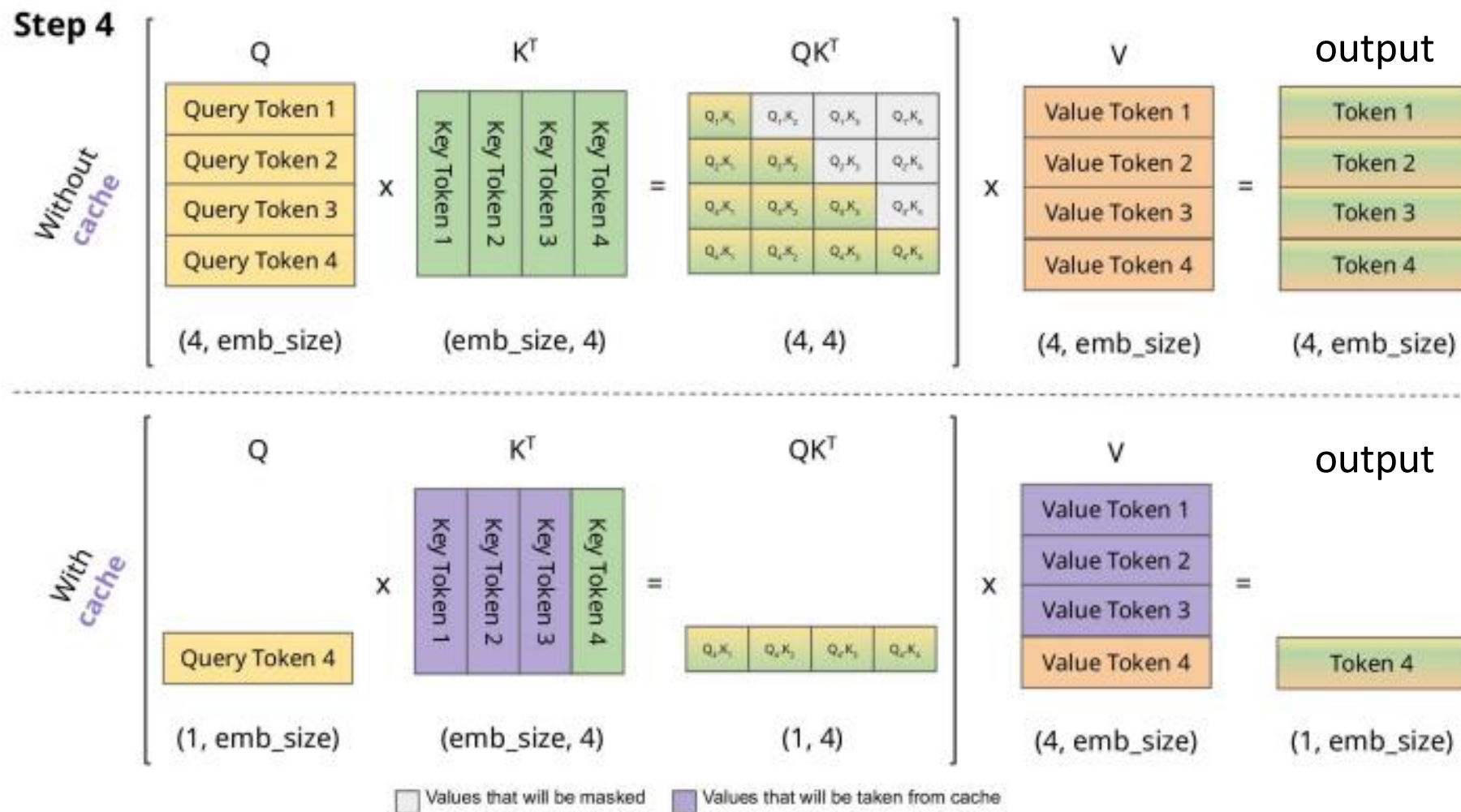
# Self-Attention with KV Cache



# Self-Attention with KV Cache



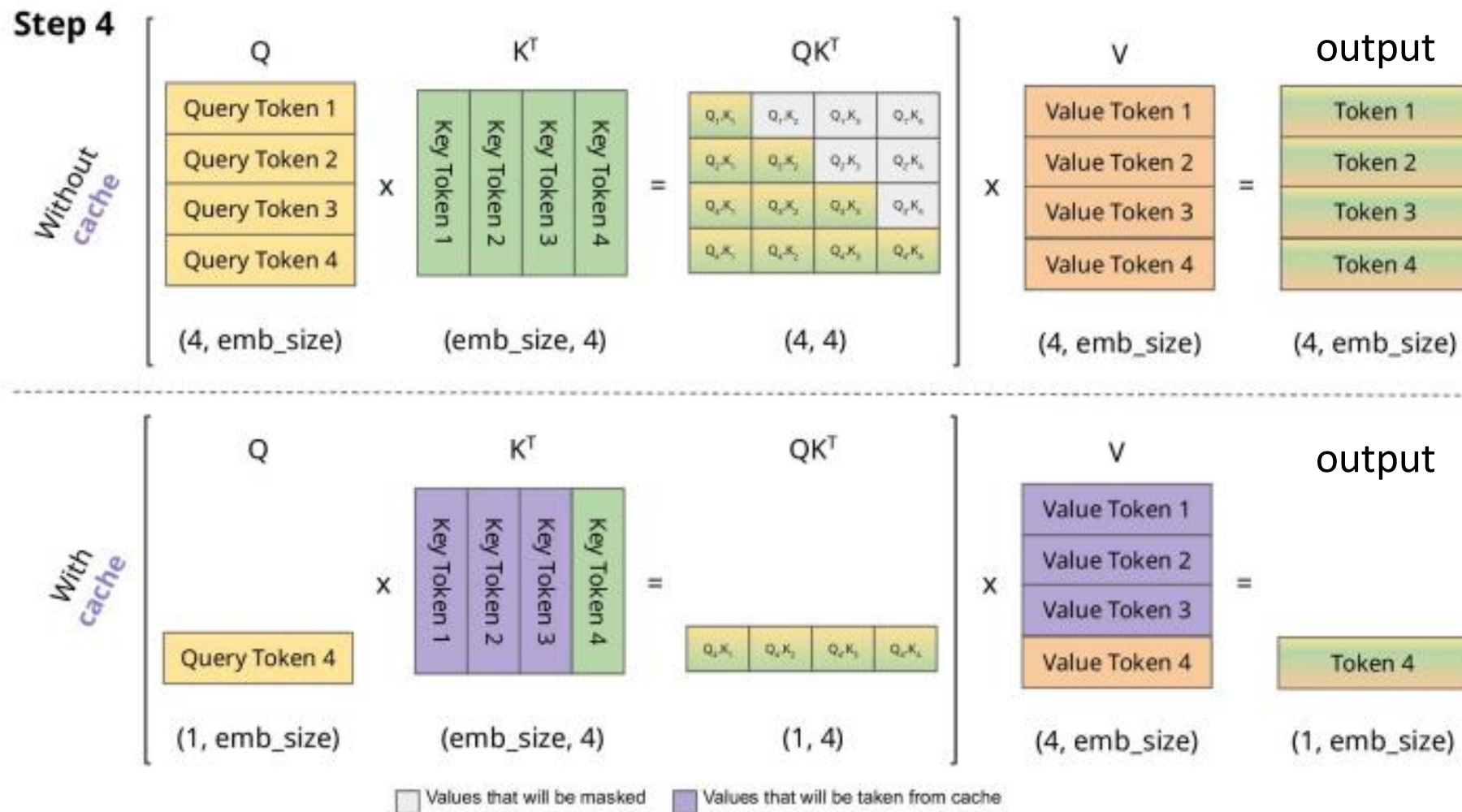
# Self-Attention with KV Cache



# Self-Attention with KV Cache



S: sequence length  
H: hidden dim size



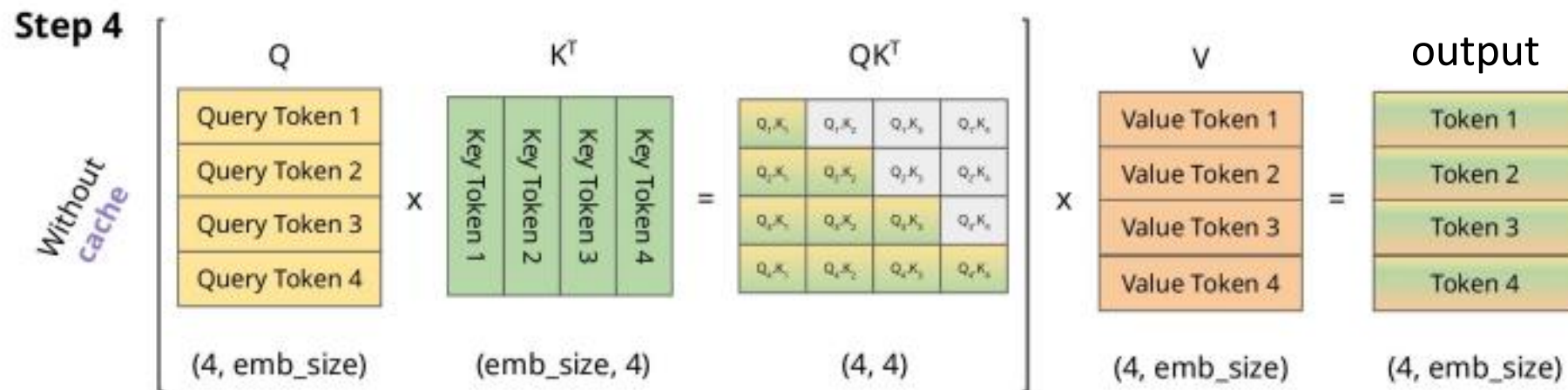
Question: Time and memory complexity without and with cache?



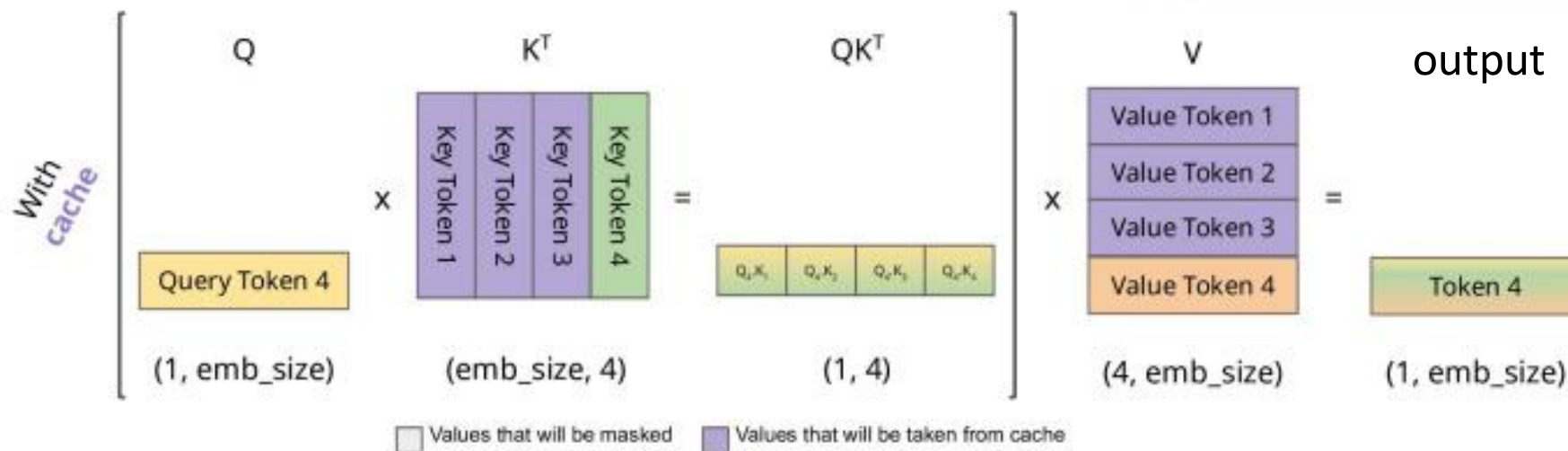
# Self-Attention with KV Cache



S: sequence length  
H: hidden dim size



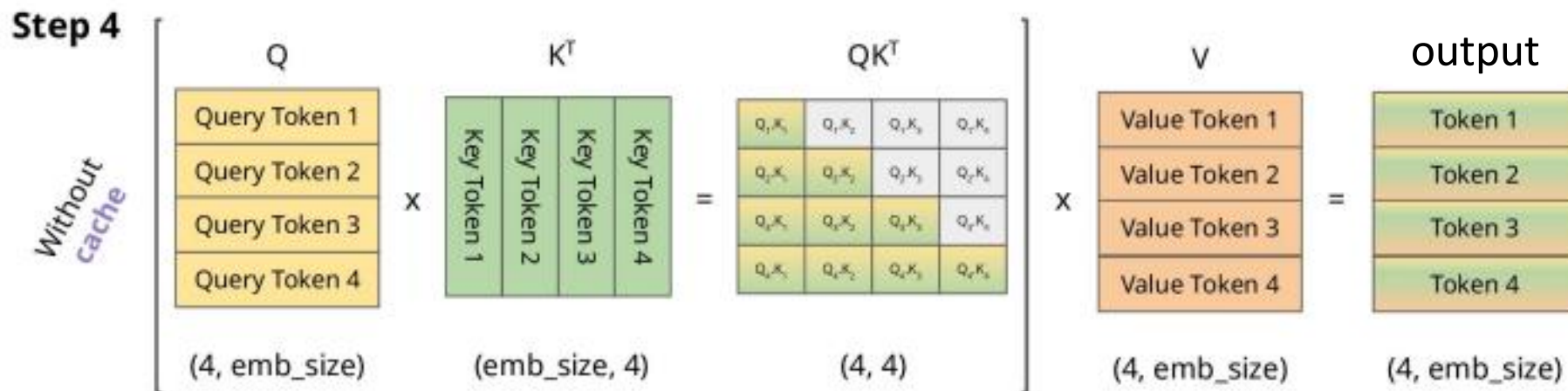
QK<sup>T</sup> Matrix size  
↓  
O(S<sup>2</sup>H) Time



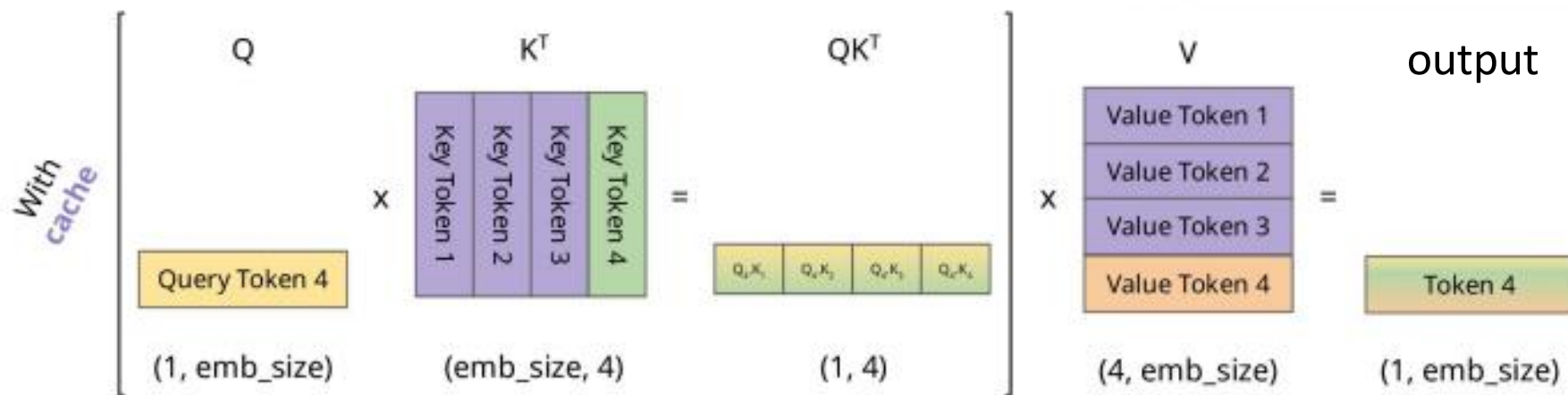
# Self-Attention with KV Cache



S: sequence length  
H: hidden dim size



$QK^T$  Matrix size  
 $\swarrow$   
 $O(S^2H)$  Time



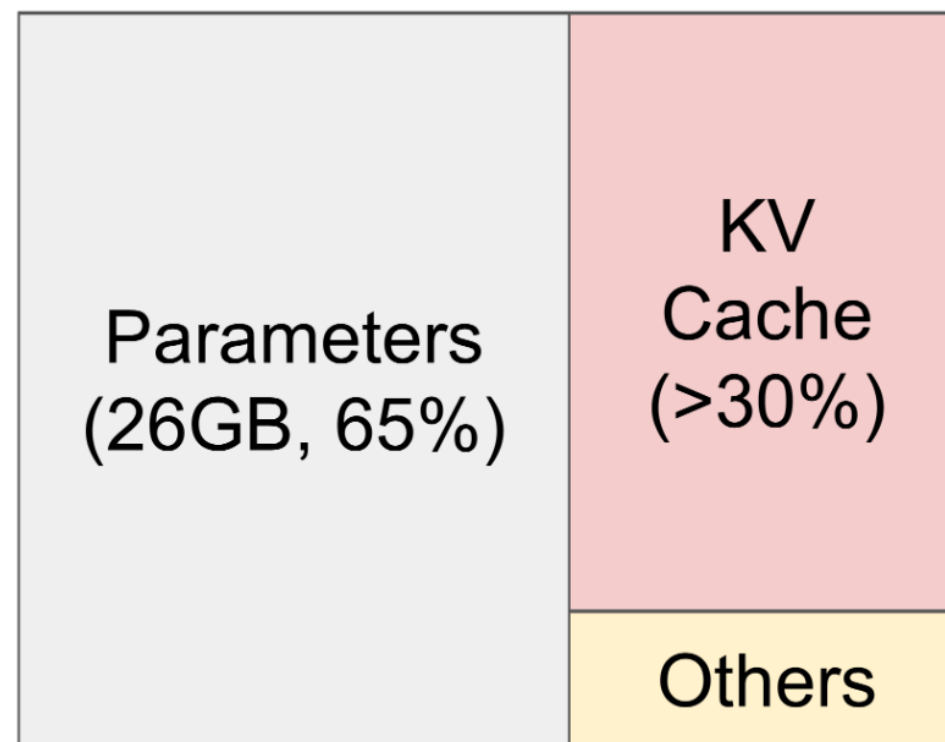
$QK^T$  row size  
 $\swarrow$   
 $O(SH)$  Time

$O(SH)$  Memory

Values that will be masked    Values that will be taken from cache

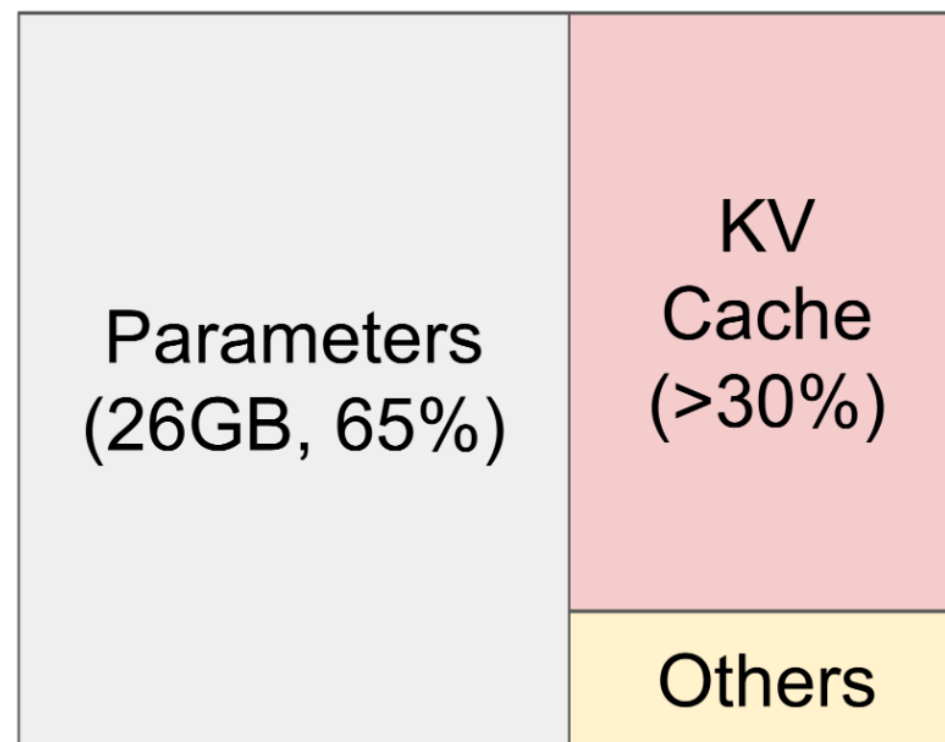


- KV cache
  - Sequence length
  - # Hidden
  - # Layers
  - Batch size



LLaMA-13B on A100 40 GB

- KV cache
  - Sequence length
  - # Hidden
  - # Layers
  - Batch size

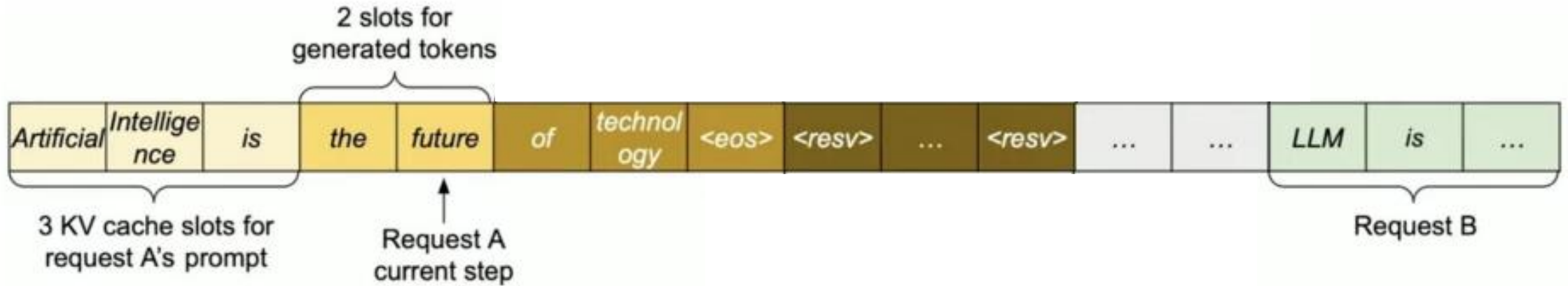


LLaMA-13B on A100 40 GB

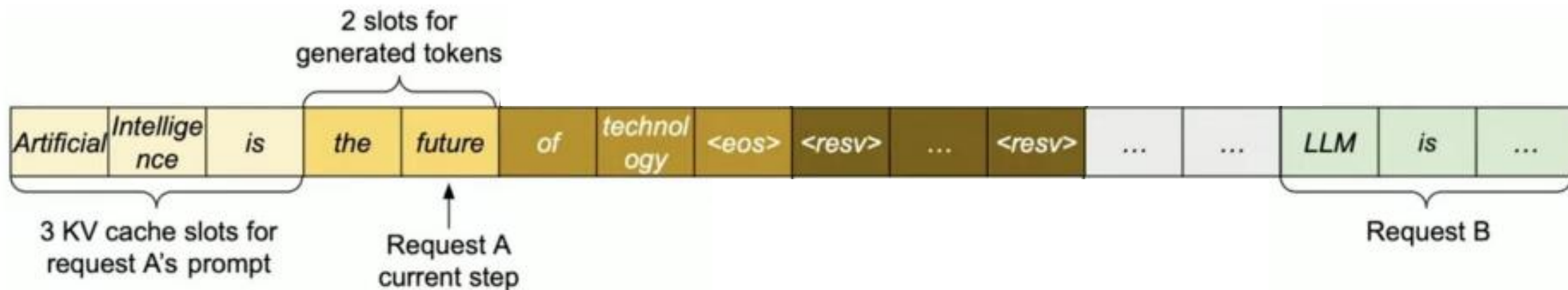
# Motivation: Memory Waste in KV Cache



How about we allocate some contiguous memory per request:  
 $\text{max-seq-len} * \text{hidden-dim} * \text{num-layers}$ ?

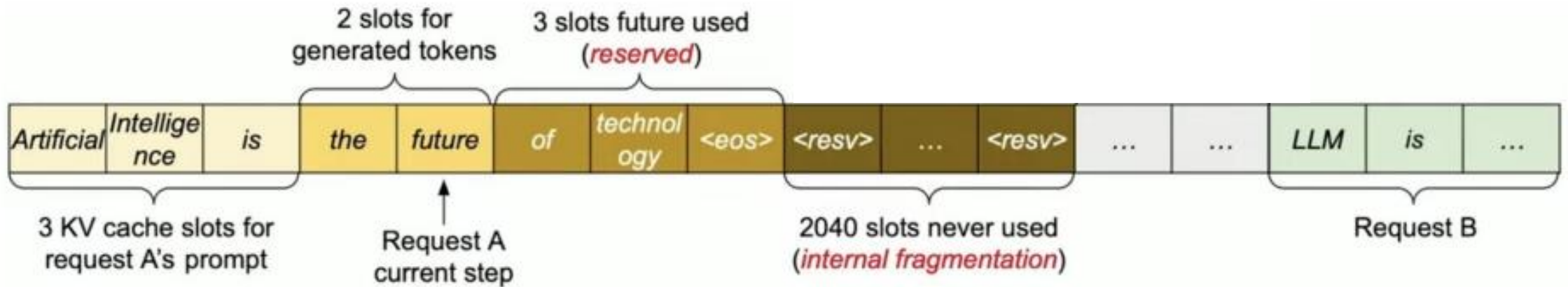


# Motivation: Memory Waste in KV Cache



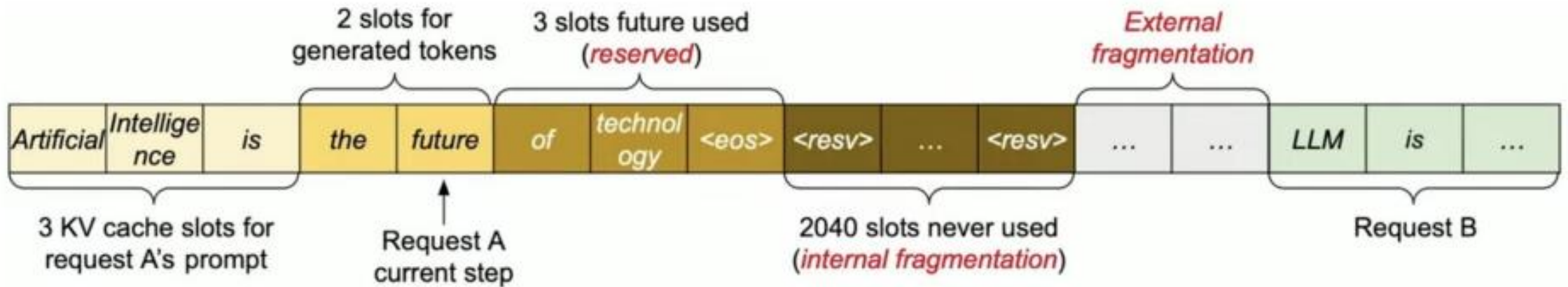
Question: Where is memory being wasted?

# Motivation: Memory Waste in KV Cache



- **Internal fragmentation:** memory reserved for a request but left unused because the final output length is shorter than expected
- **Reservation:** memory currently unused but reserved for future use by the same request

# Motivation: Memory Waste in KV Cache



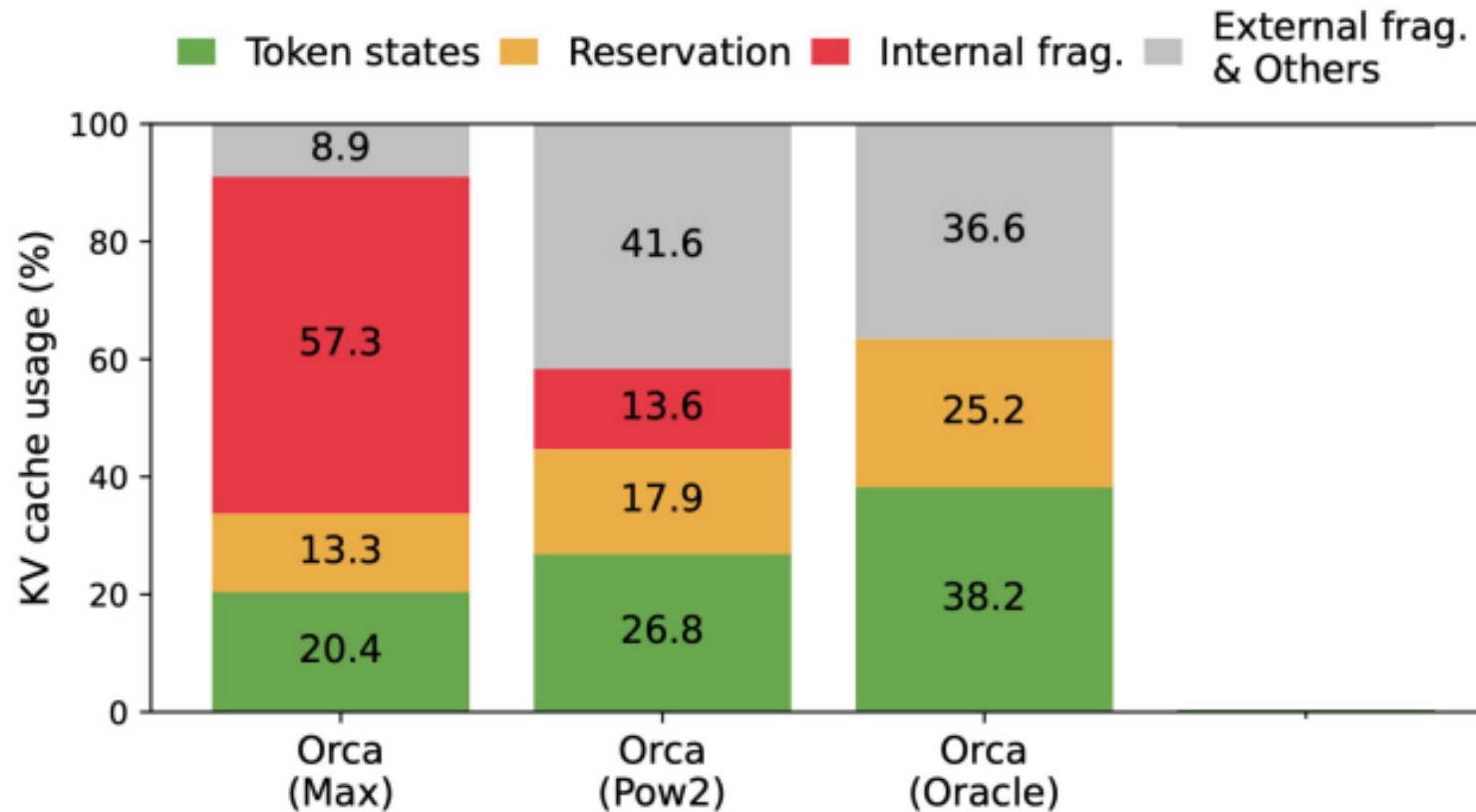
- **Internal fragmentation:** memory reserved for a request but left unused because the final output length is shorter than expected
- **Reservation:** memory currently unused but reserved for future use by the same request
- **External fragmentation:** occurs when fixed-size memory blocks do not match the variable lengths of sequences, leaving unused gaps between them that cannot be allocated to other processes



# Motivation: Memory Waste in KV cache



- Only 20-40% of KV cache is used to store the actual token states



- PagedAttention algorithm allows for storing attention key-value pairs in **non-contiguous blocks** in the memory



Key and value vectors

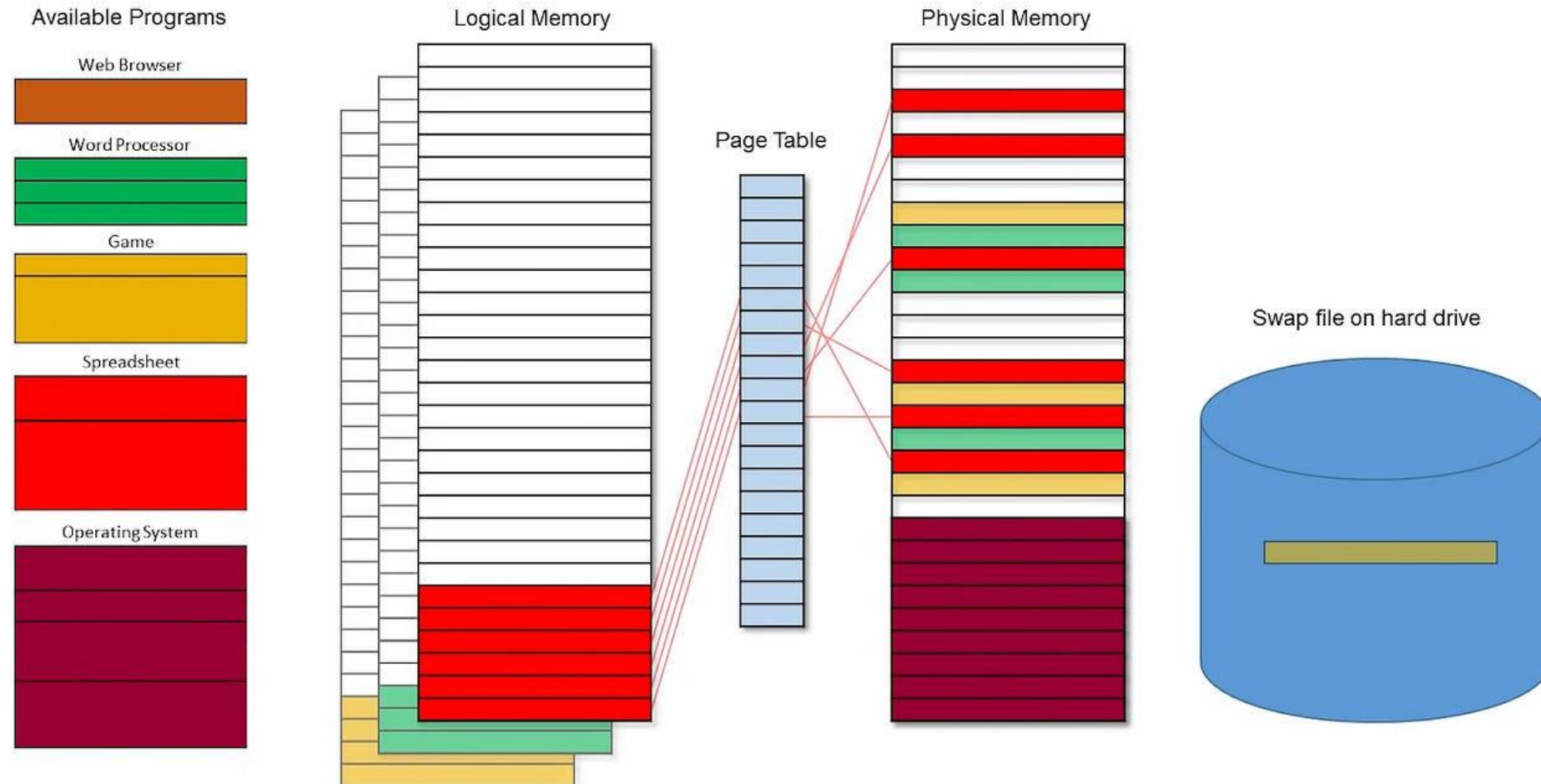
|         |         |       |     |         |
|---------|---------|-------|-----|---------|
| Block 1 | years   | ago   | our | fathers |
|         |         |       |     |         |
| Block 2 | brought | forth |     |         |
|         |         |       |     |         |
|         |         |       |     |         |
| Block 0 | Four    | score | and | seven   |

block size = 4

# Inspiration: Memory Paging (Look-up Table) in OS



## Paged Memory



- PagedAttention algorithm allows for storing attention key-value pairs in **non-contiguous blocks** in the memory
- Each KV block is a fixed-size contiguous chunk of memory that stores KV cache from left to right

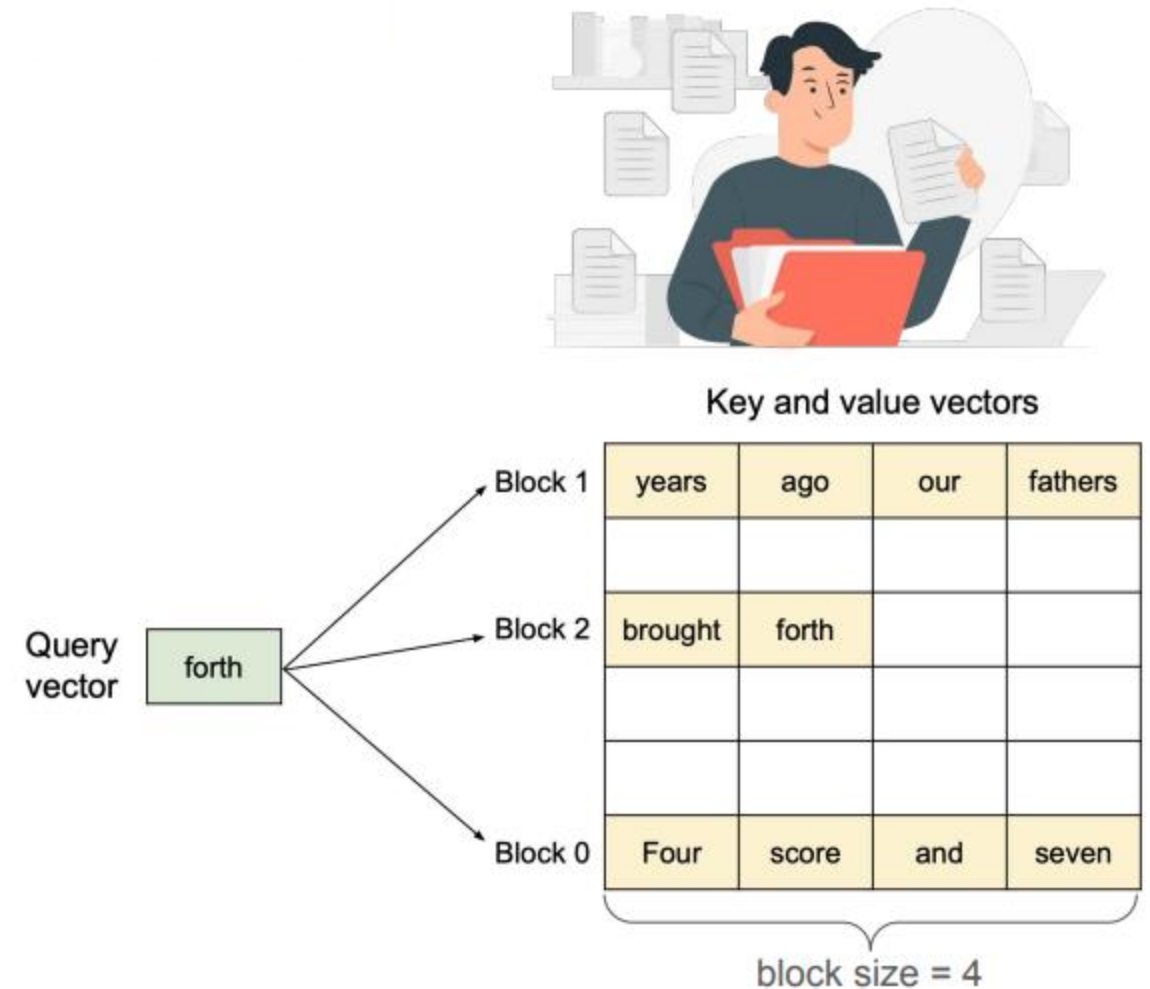


Key and value vectors

|         |         |       |     |         |
|---------|---------|-------|-----|---------|
| Block 1 | years   | ago   | our | fathers |
|         |         |       |     |         |
| Block 2 | brought | forth |     |         |
|         |         |       |     |         |
|         |         |       |     |         |
| Block 0 | Four    | score | and | seven   |

block size = 4

- PagedAttention algorithm allows for storing attention key-value pairs in **non-contiguous blocks** in the memory
- Each KV block is a fixed-size contiguous chunk of memory that stores KV cache from left to right
- Attention for each new token is computed across all tokens in the non-contiguous KV block



# Block Table For Managing KV Cache Blocks



0. Before generation.

Seq A **Prompt:** "Alan Turing is a computer scientist"  
**Completion:** ""

Logical KV cache blocks

|         |  |  |  |  |
|---------|--|--|--|--|
| Block 0 |  |  |  |  |
| Block 1 |  |  |  |  |
| Block 2 |  |  |  |  |
| Block 3 |  |  |  |  |

Block table

| Physical block no. | # Filled slots |
|--------------------|----------------|
| —                  | —              |
| —                  | —              |
| —                  | —              |
| —                  | —              |

Physical KV cache blocks

|         |  |  |  |  |
|---------|--|--|--|--|
| Block 0 |  |  |  |  |
| Block 1 |  |  |  |  |
| Block 2 |  |  |  |  |
| Block 3 |  |  |  |  |
| Block 4 |  |  |  |  |
| Block 5 |  |  |  |  |
| Block 6 |  |  |  |  |
| Block 7 |  |  |  |  |



# Block Table For Managing KV Cache Blocks



1. Allocate space and store the prompt's KV cache.

Seq  
A

**Prompt:** "Alan Turing is a computer scientist"  
**Completion:** ""

Logical KV cache blocks

|         |          |           |    |   |
|---------|----------|-----------|----|---|
| Block 0 | Alan     | Turing    | is | a |
| Block 1 | computer | scientist |    |   |
| Block 2 |          |           |    |   |
| Block 3 |          |           |    |   |

Block table

| Physical block no. | # Filled slots |
|--------------------|----------------|
| 7                  | 4              |
| 1                  | 2              |
| —                  | —              |
| —                  | —              |

Physical KV cache blocks

|         |          |           |    |   |
|---------|----------|-----------|----|---|
| Block 0 |          |           |    |   |
| Block 1 | computer | scientist |    |   |
| Block 2 |          |           |    |   |
| Block 3 |          |           |    |   |
| Block 4 |          |           |    |   |
| Block 5 |          |           |    |   |
| Block 6 |          |           |    |   |
| Block 7 | Alan     | Turing    | is | a |

# Block Table For Managing KV Cache Blocks



2. Generated 1st token.

Seq  
A

**Prompt:** "Alan Turing is a computer scientist"  
**Completion:** "and"

Logical KV cache blocks

|         |          |           |     |   |
|---------|----------|-----------|-----|---|
| Block 0 | Alan     | Turing    | is  | a |
| Block 1 | computer | scientist | and |   |
| Block 2 |          |           |     |   |
| Block 3 |          |           |     |   |

Block table

| Physical block no. | # Filled slots |
|--------------------|----------------|
| 7                  | 4              |
| 1                  | 3              |
| —                  | —              |
| —                  | —              |

Physical KV cache blocks

|         |          |           |     |   |
|---------|----------|-----------|-----|---|
| Block 0 |          |           |     |   |
| Block 1 | computer | scientist | and |   |
| Block 2 |          |           |     |   |
| Block 3 |          |           |     |   |
| Block 4 |          |           |     |   |
| Block 5 |          |           |     |   |
| Block 6 |          |           |     |   |
| Block 7 | Alan     | Turing    | is  | a |

# Block Table For Managing KV Cache Blocks



## 3. Generated 2nd token.

Seq A Prompt: "Alan Turing is a computer scientist"  
Completion: "and mathematician"

Logical KV cache blocks

|         |          |           |     |                    |
|---------|----------|-----------|-----|--------------------|
| Block 0 | Alan     | Turing    | is  | a                  |
| Block 1 | computer | scientist | and | mathe-<br>matician |
| Block 2 |          |           |     |                    |
| Block 3 |          |           |     |                    |

Block table

| Physical block no. | # Filled slots |
|--------------------|----------------|
| 7                  | 4              |
| 1                  | 4              |
| —                  | —              |
| —                  | —              |

Physical KV cache blocks

|         |          |           |     |                    |
|---------|----------|-----------|-----|--------------------|
| Block 0 |          |           |     |                    |
| Block 1 | computer | scientist | and | mathe-<br>matician |
| Block 2 |          |           |     |                    |
| Block 3 |          |           |     |                    |
| Block 4 |          |           |     |                    |
| Block 5 |          |           |     |                    |
| Block 6 |          |           |     |                    |
| Block 7 | Alan     | Turing    | is  | a                  |

# Block Table For Managing KV Cache Blocks



4. Generated 3rd token. Allocate new block.

Seq A Prompt: "Alan Turing is a computer scientist"  
Completion: "and mathematician renowned"

Logical KV cache blocks

|         |          |           |     |                    |
|---------|----------|-----------|-----|--------------------|
| Block 0 | Alan     | Turing    | is  | a                  |
| Block 1 | computer | scientist | and | mathe-<br>matician |
| Block 2 | renowned |           |     |                    |
| Block 3 |          |           |     |                    |

Block table

| Physical<br>block no. | # Filled<br>slots |
|-----------------------|-------------------|
| 7                     | 4                 |
| 1                     | 4                 |
| 3                     | 1                 |
| —                     | —                 |

Physical KV cache blocks

|         |                     |           |     |                    |
|---------|---------------------|-----------|-----|--------------------|
| Block 0 |                     |           |     |                    |
| Block 1 | computer            | scientist | and | mathe-<br>matician |
| Block 2 | Allocated on demand |           |     |                    |
| Block 3 | renowned            |           |     |                    |
| Block 4 |                     |           |     |                    |
| Block 5 |                     |           |     |                    |
| Block 6 |                     |           |     |                    |
| Block 7 | Alan                | Turing    | is  | a                  |

# Block Table For Managing KV Cache Blocks



5. Generated 4th token.

Seq A Prompt: "Alan Turing is a computer scientist"  
Completion: "and mathematician renowned for"

Logical KV cache blocks

|         |          |           |     |                    |
|---------|----------|-----------|-----|--------------------|
| Block 0 | Alan     | Turing    | is  | a                  |
| Block 1 | computer | scientist | and | mathe-<br>matician |
| Block 2 | renowned | for       |     |                    |
| Block 3 |          |           |     |                    |

Block table

| Physical block no. | # Filled slots |
|--------------------|----------------|
| 7                  | 4              |
| 1                  | 4              |
| 3                  | 2              |
| -                  | -              |

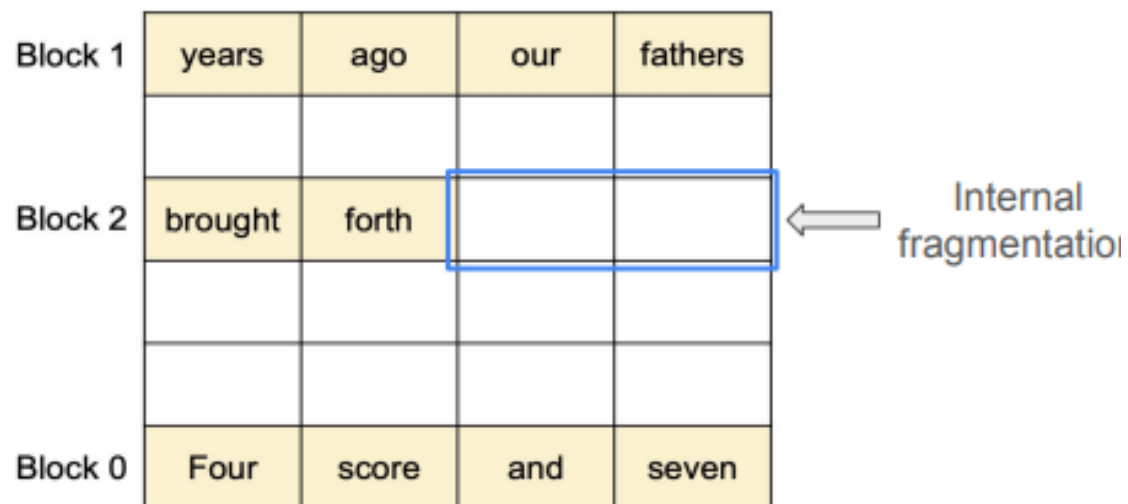
Physical KV cache blocks

|         |          |           |     |                    |
|---------|----------|-----------|-----|--------------------|
| Block 0 |          |           |     |                    |
| Block 1 | computer | scientist | and | mathe-<br>matician |
| Block 2 |          |           |     |                    |
| Block 3 | renowned | for       |     |                    |
| Block 4 |          |           |     |                    |
| Block 5 |          |           |     |                    |
| Block 6 |          |           |     |                    |
| Block 7 | Alan     | Turing    | is  | a                  |

# Memory Efficiency of PagedAttention



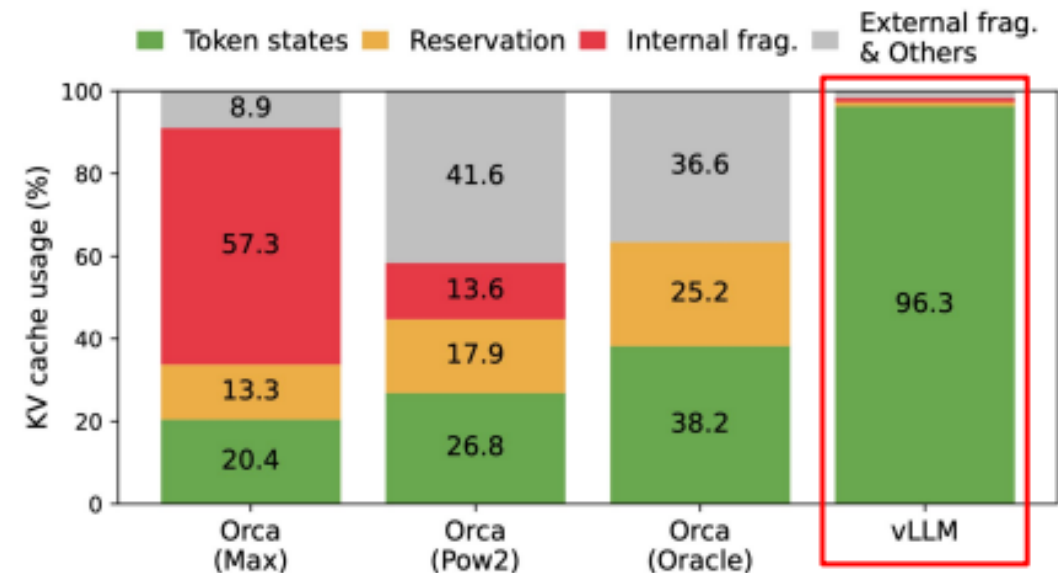
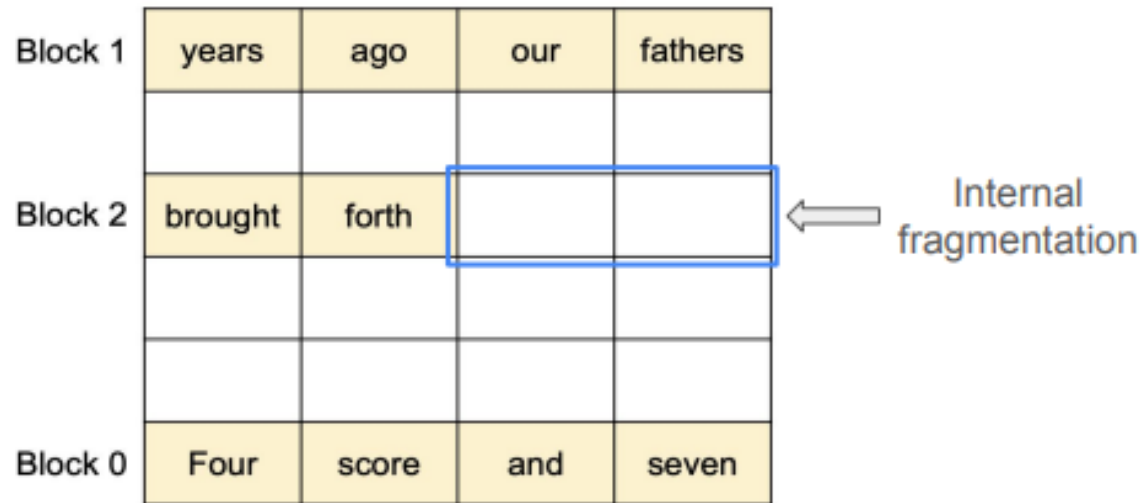
- Minimal internal fragmentation
  - # of wasted tokens per sequence < block size



# Memory Efficiency of PagedAttention



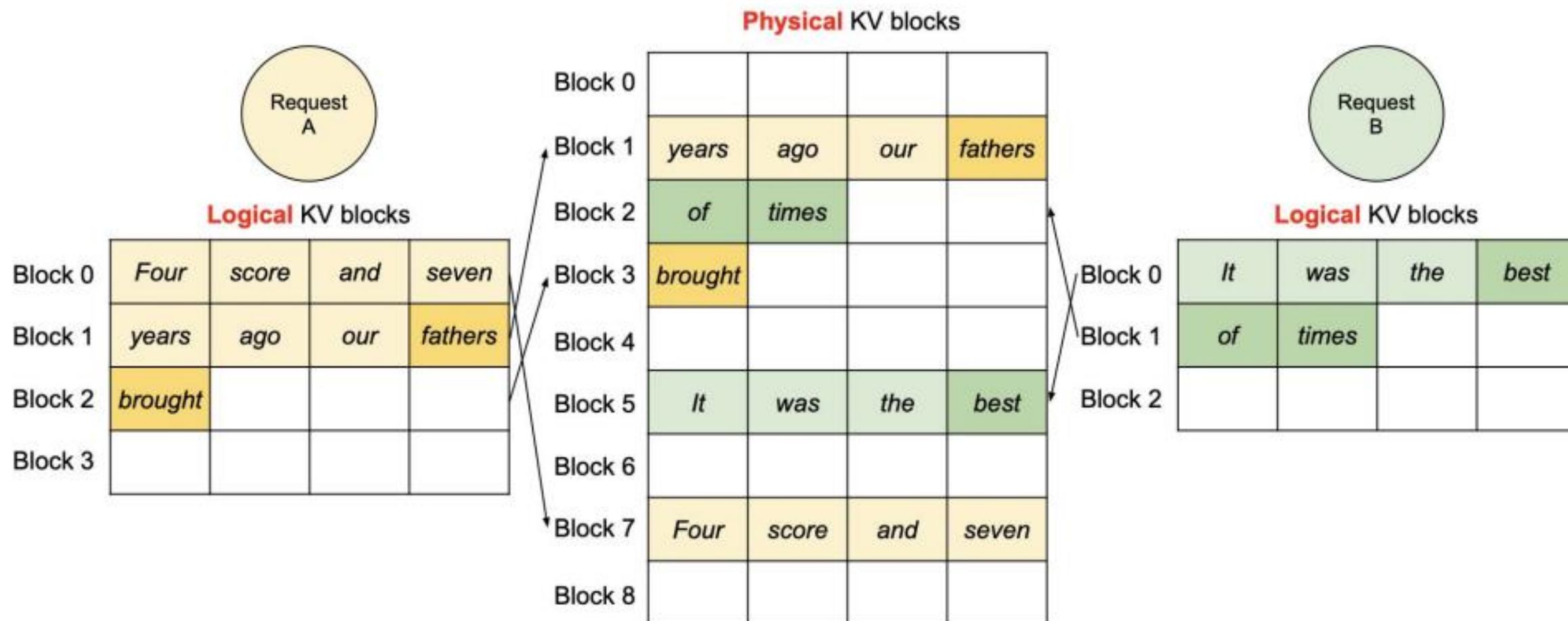
- Minimal internal fragmentation
  - # of wasted tokens per sequence < block size
- No external fragmentation (as we do not overclaim blocks)



- With PagedAttention, wasted KV cache space is < 4% (3-5x improved memory utilization)



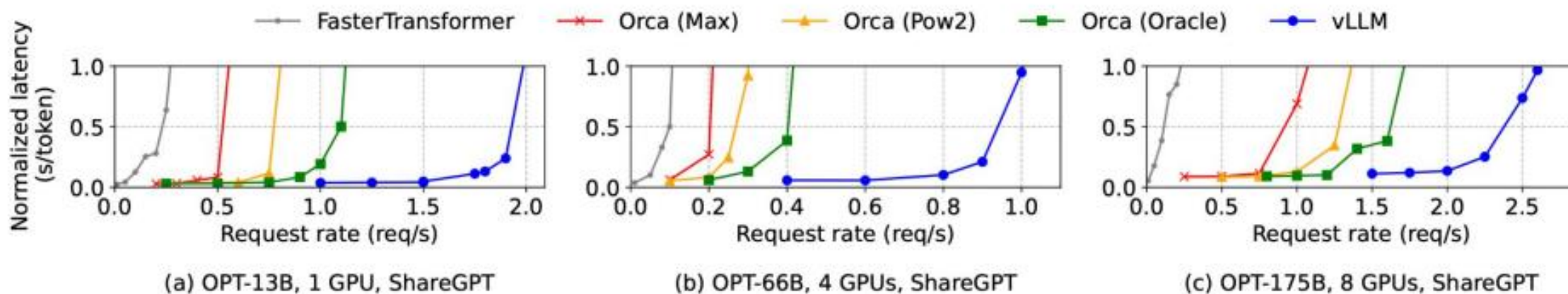
# Handling Multiple Requests



# vLLM Performance with Basic Generation



- one sample per request on three models and two datasets



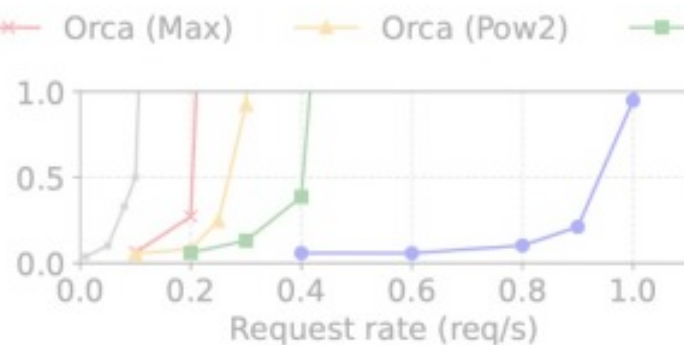
# vLLM Performance with Basic Generation



- one sample per request on three models and two datasets



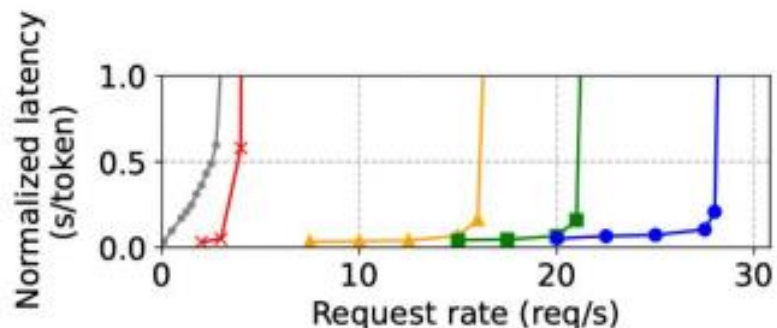
(a) OPT-13B, 1 GPU, ShareGPT



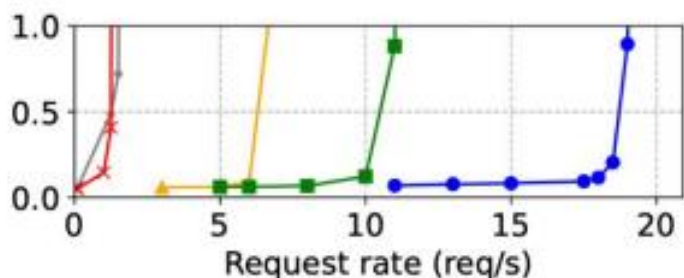
(b) OPT-66B, 4 GPUs, ShareGPT



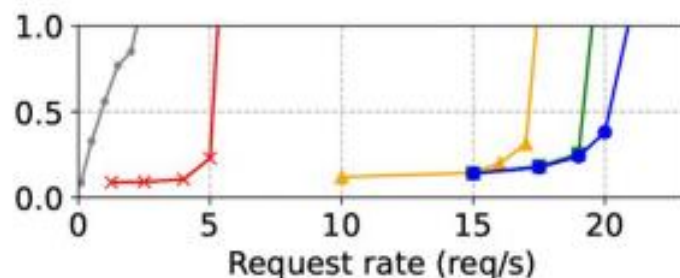
(c) OPT-175B, 8 GPUs, ShareGPT



(d) OPT-13B, 1 GPU, Alpaca



(e) OPT-66B, 4 GPUs, Alpaca



(f) OPT-175B, 8 GPUs, Alpaca

- Reduces memory fragmentation with paging
- Reduces memory usage with KV block sharing
- Enables batching of more requests, increasing the throughput of LLM inference

4-page Mid-semester Report Due on Nov. 07

## SGLang: Efficient Execution of Structured Language Model Programs

Lianmin Zheng<sup>2\*</sup> Liangsheng Yin<sup>3</sup> Zhiqiang Xie<sup>1</sup> Chuyue Sun<sup>1</sup> Jeff Huang<sup>4</sup>  
Cody Hao Yu<sup>5</sup> Shiyi Cao<sup>2</sup> Christos Kozyrakis<sup>1</sup> Ion Stoica<sup>2</sup> Joseph E. Gonzalez<sup>2</sup>  
Clark Barrett<sup>1</sup> Ying Sheng<sup>1\*</sup>

<sup>1</sup> Stanford University   <sup>2</sup> UC Berkeley   <sup>3</sup> Shanghai Jiao Tong University  
<sup>4</sup> Texas A&M University   <sup>5</sup> Independent Researcher

Allow prompts sharing the prefix input to use the same KV cache (~trie tree)

